

Name: _____

TRANSFORMATIONS, (ROTATION)

Date: _____

LO: I can rotate shapes about a given centre by 90, 180, or 270 degrees.

Rotation

A **rotation** turns a shape around a fixed point called the **centre of rotation**. You need to know the centre, angle, and direction (clockwise or anticlockwise).

Key rules: 90° clockwise: (x,y) (y,-x). 90° anticlockwise: (x,y) (-y,x). 180°: (x,y) (-x,-y) about the origin.

Quick examples

●	(3, 1) rotated 90° clockwise about origin = (1, -3)	(x,y) (y,-x)
●	(3, 1) rotated 180° about origin = (-3, -1)	negate both
●	Rotation preserves shape and size	congruent image
●	90° clockwise is the same as 90° anticlockwise	they are opposite directions

Key idea

To describe a rotation, state: centre, angle, and direction.
The shape does not change size.

90° CW about O: (x, y)

(y, -x)

180°: (x, y) (-x, -y)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – 90 clockwise about origin

Rotate (2, 5) by 90° clockwise about (0, 0).

Rule: (x, y) (y, -x)

(2, 5) (5, -2)

Image is at (5, -2)

(5, -2)

Swap coordinates and negate the new y.

Example 2 – 180 about origin

Rotate (3, -1) by 180° about (0, 0).

Rule: (x, y) (-x, -y)

(3, -1) (-3, 1)

Image is at (-3, 1)

(-3, 1)

Negate both coordinates.

Example 3 – 90 anticlockwise about origin

Rotate (4, 2) by 90° anticlockwise about (0, 0).

Rule: (x, y) (-y, x)

(4, 2) (-2, 4)

Image is at (-2, 4)

(-2, 4)

Swap and negate the new x.

Example 4 – rotation about a different centre

Rotate (5, 2) by 180° about (3, 1).

Translate to origin: (5-3, 2-1) = (2, 1)

Apply 180°: (-2, -1)

Translate back: (-2+3, -1+1) = (1, 0)

(1, 0)

Translate to origin, rotate, translate back.

YOUR TURN – PRACTICE (deliberate practice)

1. 90° clockwise about origin

Rotate each point 90° clockwise about (0,0).

- a) (1, 3)
- b) (4, 2)
- c) (-1, 5)
- d) (0, 4)

2. 180° about origin

Rotate each point 180° about (0,0).

- a) (3, 2)
- b) (-1, 4)
- c) (5, -3)
- d) (-2, -6)

3. 90° anticlockwise about origin

Rotate each point 90° anticlockwise about (0,0).

- a) (2, 3)
- b) (5, 1)
- c) (-3, 4)
- d) (6, 0)

4. About a non-origin centre

Rotate 180° about the given centre.

- a) (4, 3) about (2, 2)
- b) (5, 1) about (3, 3)
- c) (1, 4) about (3, 2)
- d) (6, 5) about (4, 3)

5. Mixed rotations

Perform each rotation.

- a) (3, 1) 90° CW about origin
- b) (2, 4) 180° about origin
- c) (1, 5) 90° ACW about origin
- d) (4, 2) 90° CW about (2, 1)
- e) (1, 3) 180° about (2, 2)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 180° rotation about origin:
(x,y) becomes (-x,-y) ☐

Correct answer: _____

Reason: _____

b) 90° clockwise and 90° anticlockwise give the same image ☐

Correct answer: _____

Reason: _____

c) Rotation preserves the shape and size ☐

Correct answer: _____

Reason: _____

d) A 360° rotation returns the shape to a different position ☐

Correct answer: _____

Reason: _____

e) The centre of rotation does not move ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Point A(3, 1) is rotated 90° clockwise about (1, 2) to get A'. Find A'. Then rotate A' by 90° clockwise about the same centre. What single rotation maps A to A'?

Expression: _____

Simplified: _____

b) A square has vertices at (1,1), (3,1), (3,3), (1,3). It is rotated 90° anticlockwise about (2,2). Find the image vertices and verify the image is still a square.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

